Paper 68 — Special Properties of Gradient Descent with Large Learning Rates

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The paper in one minute

Question: Can the *step size itself*, even without SGD noise, change which minimum GD eventually reaches?
Model/Method: Compare large- and small-step full-batch GD in experiments and analyze a structured class of smooth nonconvex functions with sharp spurious minima.
Finding: A sufficiently large step may make a sharp local basin repelling; under additional global landscape conditions, GD instead reaches the global minimum.
Why we care: A learning rate does more than control speed: it can change the chosen optimization trajectory and final basin.

1. Start with a non-global minimum

Suppose a loss has a sharp, suboptimal valley and a broader, lower valley. At small step sizes, GD can settle into the sharp bad valley. A larger step can overshoot this valley repeatedly and move into a region leading toward the global solution. This mechanism is *deterministic*: mini-batch noise is not needed to produce it.
The paper emphasizes that the usual smooth-GD theory studies small η where loss decreases reliably each step. That regime cannot explain how step size can change which basin is reached.

2. Worked example: local instability can reject a valley

Near a local minimizer $x^\dagger$, approximate the loss by $L(x) \approx L(x^\dagger) + \frac{\kappa}{2}(x - x^\dagger)^2$, with $\kappa > 0$. If $e_t = x_t - x^\dagger$, then

$$e_{t+1} \approx (1 - \eta\kappa)e_t.$$

Take a sharp valley with $\kappa = 10$. With $\eta = 0.1$, $e_{t+1} = 0$ in this quadratic approximation: it is strongly attracting. With $\eta = 0.25$, $e_{t+1} = -1.5e_t$: alternating errors grow, so the same point becomes locally repelling.
Critical warning. This only proves local instability of the *sharp valley*. It does not prove that the iterate will avoid every other poor region or reach a global minimum. That stronger conclusion requires the global structural assumptions of the paper's Theorem 1.

3. What the main theorem actually assumes

Theorem 1 constructs a smooth function class with a sharp local minimum $x^\dagger$ in a special region M and a global minimum $x^\star$. Outside M, an *one-point strong-convexity-type* condition directs progress toward $x^\star$. Near $x^\dagger$, local curvature and separation assumptions permit a sufficiently large step to escape. Its point is a contrast:

$$\begin{aligned} \text{small step:} &\quad \text{convergence to } x^\dagger \text{ from a local basin,} \\ \text{large step:} &\quad \text{convergence to } x^\star \text{ under stronger global conditions.} \end{aligned}$$

This is not an all-purpose global convergence theorem for arbitrary neural networks. In particular, escaping a bad basin alone is logically weaker than proving eventual convergence to the best basin.

Why the theorem is a genuine contribution

The authors isolate a trajectory-selection effect of *discretization*. Gradient flow and very small-step GD have different basin behavior from sufficiently large-step GD; stochastic noise cannot fully explain the effect.

4. Evidence and interpretation

The paper's experiments include full-batch and stochastic training comparisons that vary learning rate while examining which solutions are reached. The deterministic observations support the argument that noise alone does not explain all advantages of large steps. Figures 1, 3 and 4 illustrate overshooting/avoidance of narrow valleys and the formal neighborhood construction.

A subtle point: If η is excessively large, the entire algorithm may diverge. The paper advocates analyzing *intermediate* large-step regimes, not choosing the largest imaginable rate.

5. Proof structure and experimental interpretation

The local escape lemma uses sharp-basin curvature to show that a large enough step cannot remain nearby indefinitely. The *global* theorem adds structure outside that exceptional region, so excursions ultimately lead toward the global solution. These are logically separate: *repulsion from a bad basin* and *attraction toward the good solution*.

When interpreting plots, compare full-batch GD with stochastic training at matched step sizes. Evidence that full-batch GD benefits from large steps rules out an explanation requiring gradient sampling noise. It does *not* rule out additional SGD-noise effects in other settings or imply that arbitrarily large rates work.

5. Assumptions and limitations

Assumptions: A globally smooth function; one-point strong-convexity structure relative to $x^\star$ outside a special bad-minimum region; local curvature and geometric separation around $x^\dagger$; suitable step-size interval and initialization distribution.
Limits: The theorem's function class is highly structured. Full-batch neural-network experiments cannot establish the universal frequency of this effect, and large steps can cause divergence or poorer performance in other settings. Escaping a sharp minimum is not identical to escaping a strict saddle.

6. Connections to the survey

Safran–Shamir (2018) establishes that spurious minima can exist; large-step GD offers a *possible* dynamical way to avoid selected sharp ones. **Li–Wang–Li (2022)** emphasizes curvature oscillation, and **Chen–Bruna (2023)** studies period-two behavior instead of alternative basin selection.

7. Quick reference

Concept	Meaning / equation
Local error multiplier	$e_{t+1} \approx (1-\eta\kappa)e_t$
Sharp-basin instability	$\eta\kappa > 2$
Main contrast	small-step local vs. large-step global basin
Global theorem ingredient	one-point convexity outside bad region
Original to read	Sections 3–5; Lemma 1, Theorem 1

8. Papers Master note

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Shows large learning rates can change deterministic GD basin selection, in both experiments and a specially structured smooth nonconvex function class. Local repulsion from a sharp spurious minimum is not a general global-optimality guarantee.

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